

Name: K RAJ BHARATH K RAJ BHARATH | Reg. No: RA2511003011548

Unit – 2 : Session – 7 : SLO – 2 | Implementation of Linked Lists — Concepts

■ I. Fill in the Blanks — Linked List Basics

- A linked list is a collection of NODES, where each node contains data and a pointer to the NEXT NODE.
- In C, the keyword STRUCT is used to define a node in a linked list.
- The pointer used to point to the first node of the linked list is called the HEAD (START).
- The pointer in the last node of a singly linked list points to NULL.
- A node in a linked list typically contains two fields: DATA and NEXT (LINK).
- Dynamic memory allocation for a new node is done using the MALLOC() function.

■ II. Fill in the Blanks — Traversal, Insertion and Deletion

- To traverse a linked list, a TEMPORARY (PTR) pointer is used to iterate through each node.
- To insert a node at the beginning of the linked list, update the new node's link to point to the CURRENT HEAD (START).
- After inserting a node at the beginning, the HEAD (START) must be updated to point to the new node.
- To delete a node, it's important to FREE the memory after unlinking the node.
- A NULL check should always be done before accessing a node's data through a pointer.
- Inserting a node at the end requires traversal until the node's NEXT pointer is NULL.